

**TOPIC FOUR
MOLECULAR GENETICS / MODERN
BIOCHEMISTRY.**

Lecture 6

INTRODUCTION

- This subject teaches us that the genetic information is stored in DNA or RNA (RNA viruses). And it teaches us the central dogma; that is the flow of genetic information from DNA to DNA from DNA to RNA, and then to protein. Human beings are highly developed species.
- How a person looks, behaves, suffers from diseases and the like are dictated by genetic material called DNA. The information is inherited from parent to offspring. The same is replicated from parent DNA to daughter DNA.
- First RNA is synthesized from DNA (Transcription) which is translated in to Proteins.
- These proteins are responsible for various metabolic functions. Protein expression during development, adaptation, aging and other related processes of life are controlled by DNA.
- Changes in the genetic material cause **hereditary diseases**.

NUCLEIC ACIDS

- Nucleic acids are present in nucleus and Mitochondria
- They are found in two basic structural forms, deoxyribonucleic acid (DNA) and ribonucleic acid(RNA).

THE STRUCTURE OF DNA

- Deoxyribonucleic acid (DNA) is polymers of deoxyribonucleotides attached to each other by phosphodiester linkages.
- Each deoxyribonucleotide is composed of deoxyribonucleoside & inorganic phosphate group.
- Each deoxyribonucleoside is composed of nitrogen bases and a sugar deoxyribose.
- The nitrogenous bases are purines and pyrimidines.

- The two Purine bases are **adenine** and **guanine**. They are derived from their parental compound.
- The three Pyrimidine bases are **cytosine**, **thymine** and **uracil**.

NB: Thymine is found in DNA and **uracil** is found in RNA but the other above mentioned bases are found in both DNA and RNA.

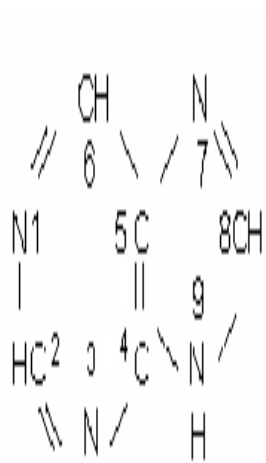
The Structure of RNA

- The building unit of RNA is ribonucleotide. Ribonucleotide differs from deoxyribonucleotide in that ribonucleotide contains “O” in the carbon 2’ sugar ribose.
- Uracil is found in RNA while Thymine is found in DNA.
- The nuclear DNA is in secondary structure, but RNA is the primary structure. Only t-RNA after post transcriptional process can be changed to tertiary structure.

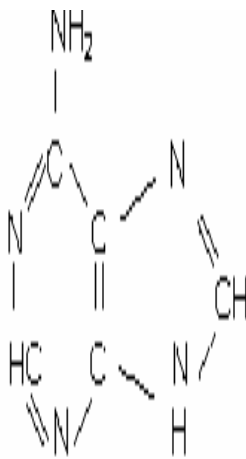
STRUCTURE OF DNA & RNA

THE STRUCTURE OF DNA

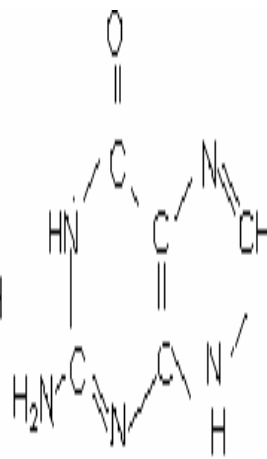
The Structure of RNA



The parental compound

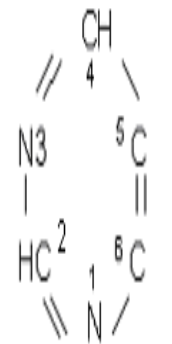


Adenine

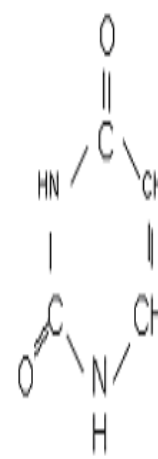


Guanine

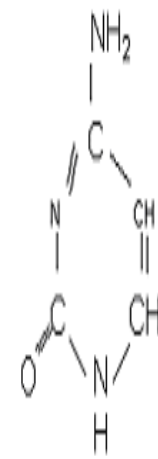
Derivatives



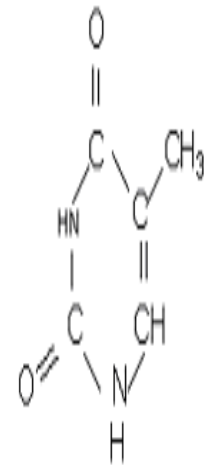
Parental compound



uracil



Cytosine



Thymine

Derivatives

Differences between RNA and DNA

	DNA	RNA
Uracil	Absent	Present
Sugar	Deoxyribose	Ribose
Site	Nucleus, mitochondria but never in cytosol	Nucleus, ribosome, cytosol, Nucleolus, mitochondria
Strands	Two helical strands	Single strand
Types	Major forms are A,B & Z hn RNA,& sn RNA.	t-RNA ,m-RNA, r- RNA, hn RNA, sn RNA .
Carries genetic information	-	Only m-RNA carries genetic information
	DNA can synthesize RNA by transcription	Usually RNA can't form DNA, except by reverse transcriptase.
Number of Bases	Equal	Not equal
Thymine	Present	Absent

RNAs protein synthesis

- The three RNAs protein synthesis are:that have important role in
- 1. Messenger RNA (mRNA)
- 2. Transfer RNA (tRNA)
- 3. Ribosomal RNA (rRNA)

Catabolism of Nucleic acids

- Pancreatic enzymes called nucleases hydrolyze both DNA and RNA to nucleotides in the intestinal tract.
- Nucleic acid is also hydrolyzed by lysosomal enzymes inside tissues. Here in the intestine, the nucleotide is also hydrolyzed to nucleoside and phosphoric acid.
- The nucleoside is absorbed in to blood and transported to peripheral tissues.
- Excess nitrogen bases are further degraded. Finally adenine and guanine are converted to uric acid in our body which is excreted through urine.

Catabolism of Nucleic acids

CONT....

Since uric acid has a precipitation character, excess uric acid in kidney causes kidney stone and in joints causes gout.

The degradation of pyrimidine bases are converted to:

- 1. Cytosine and uracil to ammonia, carbondioxide and beta-alanine
- 2. Thymine to NH_3 , CO_2 , H_2O and α -methyl β alanine.

RNA Synthesis

- Transcription is the process of RNA Synthesis directed by a DNA template. It occurs in three
- phases:-
 - 1. Initiation
 - 2. Elongation
 - 3. Termination

Translation or protein synthesis

- The genetic information which flows from DNA to mRNA will be translated to the universal language called protein.
- During transcription the base sequence of DNA determines the base sequence of mRNA.
- Under translation the base sequence of mRNA determines the amino acid sequences in protein.
- Translation has 4 – stages:
 - 1. Activation
 - 2. Initiation
 - 3. Elongation
 - 4. Termination

Clinical Significance

- Deoxyribonucleic acid (DNA) probes are coming into play in diagnosis of genetic disorders, infectious diseases, and cancers.
- Genetically engineered strains of bacteria containing recombinant DNA are producing valuable proteins such as insulin and growth hormone.
- Identifying and sequencing genes responsible for causing diseases, and genetic cloning technology has led to remarkable progress in understanding the relationships of genes and proteins.
- The molecular bases of several diseases, such as sickle cell anemia, and numerous inborn errors of metabolism are now known.

END

MANY THANKS